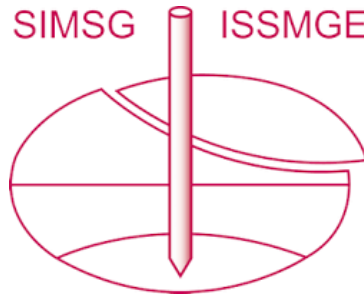


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The concept of static liquefaction and its application in the stability analysis of tailings dams

Le concept de liquéfaction statique et son application dans l'analyse de la stabilité des digues à résidus

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ABSTRACT: An extremely high seismic activity coincides in Chile with numerous tailings dams associated with the copper mining industry. In the country, there was a history of tailings dam failures due to the phenomenon of liquefaction. However, since the 70s, there have been a growing number of new large tailings dams that have had a good performance even when subjected to powerful earthquakes. The paper summarizes the instructive history of failures and successes of the newly designed tailings dams and particularly of the sand tailings dams. Additionally, a comprehensive experimental program using the conventional triaxial apparatus has been conducted on tailings materials, so the steady-state lines (or critical state lines) in the q - p - e space were obtained. These experimental data clearly show that the fines decrease the liquefaction strength of the sand tailings. Using the framework provided by the steady state of deformation permits us to understand the concept of static liquefaction and its application in the stability analysis of tailings dams directly. In the paper these concepts are applied to a documented Chilean tailings dam failure.

RÉSUMÉ: Au Chili, une activité sismique extrêmement élevée coïncide avec l'existence de nombreux barrages de rétention de résidus associés à l'industrie minière de cuivre. Au pays, il y a eu des ruptures historiques de barrages de résidus liées au phénomène de liquéfaction. Mais, depuis la décennie des années 70, un nombre croissant de nouveaux grands barrages de rétention de résidus ont eu de bonnes performances même lorsqu'ils ont été soumis à de très forts séismes. L'article résume l'histoire des échecs et des succès des barrages de rétention de résidus nouvellement conçus et, en particulier, ceux des barrages confectionnés avec des sables de résidus compactés. Un complet programme expérimental utilisant l'appareil triaxial conventionnel a été mené sur les matériaux de résidus, permis de déterminer les lignes de plasticité parfaite dans les diagrammes q - p' et $-e$ - $\log p'$. Les données expérimentales montrent clairement que la présence de fines réduit la résistance à la liquéfaction des résidus de sable. L'utilisation du cadre fourni par la plasticité parfaite permet de comprendre directement le concept de liquéfaction statique et son application dans l'analyse de la stabilité des barrages. Ces concepts sont appliqués à une rupture de barrage de rétention de résidus documentée au Chili.

KEYWORDS: Tailings dams, earthquakes, liquefaction, flow failure, static liquefaction.

1 INTRODUCTION

Mining operations produce large quantities of tailings that must be handled and stored stably in a technical frame that has to conciliate both economic restrictions and environmental sustainability. Unfortunately, it is not wrong to indicate that the history of tailings deposits is marked by episodes of catastrophic failures that have caused a significant number of victims and cost enormous material losses. Indeed this statement is confirmed by the recent catastrophic collapses of tailings dams that occurred in Canada (Mount Polley) and Brazil (Brumadinho), where the tailings that flowed downstream severely affected the environment, and in the Brazilian case, many fatalities had to be lamented.

This unfortunate empirical evidence left by the mining industry worldwide raises the question about the reasons for this adverse scenario. It is possible to answer that most of the failures can be explained by the combination of the following main factors: bad practices, operational and construction deficiencies, lack of geotechnical knowledge, uncertainties in the strength of critical materials that control the physical stability, and governance.

It is crucial to understand that a tailings dam is a mining facility that grows gradually because its function is to meet the needs of the mine in terms of storing all the generated tailings. Therefore, its construction usually takes several decades and in some cases, it coincides with the lifetime of the mine. In this context, the regular operation of the tailings deposit includes the continuous construction of the tailings dam. Thus, operation and construction cannot be separated, meaning that that difficulty in

the operation may severely affect the resulting constructed tailings dam. For example, cycloned tailings sands may have too much fines, the required amount of cycloned tailings is not produced, the compaction achieved in the dam could be low or the designed geometry could not be obtained, among others. In other words, operation-construction may create unwanted weaknesses in a tailings sand dam, which depending on the robustness of the design, could result in catastrophic failures.

Furthermore, it is not unusual a poor communication between the operators of the tailings deposit, the engineers in charge of mining production, and the different levels of the mining organization, which implies poor governance because the directors and stakeholders may not be aware of the actual risk of the tailings storage facility. As a result, proper decision-making at different levels of the mining company is limited, especially in both developed and developing countries.

On the other hand, the geotechnical knowledge needed to evaluate the physical stability of tailings sand dams, which is related to evaluating the phenomenon of liquefaction, is well developed since the pioneering work of Casagrande, Castro and Seed among others. However, this geotechnical know-how is not well understood by engineers which expertise is not explicitly associated with the liquefaction phenomenon.

Despite the serious risk that a deficient design of these structures may imply, only a few universities in the world have courses that specifically address the geotechnical topics necessary to properly carry out the characterization, design, analysis, construction and monitoring of tailings dams. In this context, in the present article, the geotechnical framework

needed to analyze the physical stability of tailings dams is explained.

2 DAM CONSTRUCTIONS OF CONVENTIONAL TSF

Although water is a limited resource in most mining countries, conventional tailings storage facilities that use a significant amount of water remain attractive due to their relatively lower cost. Nevertheless, the thickened, paste and filtered tailings deposits are gradually gaining adherents due to the greater degree of environmental sustainability that they represent.

Conventional tailings storage facilities are the ones that have experienced a collapse, and therefore, need to be analyzed in detail. A conventional tailings deposit consists of one or various tailings dams that allow the generation of a basin where the tailings are stored. For the construction of the tailings dams, there are three different procedures referred to as upstream, downstream, and centerlines, which are illustrated in Fig. 1.

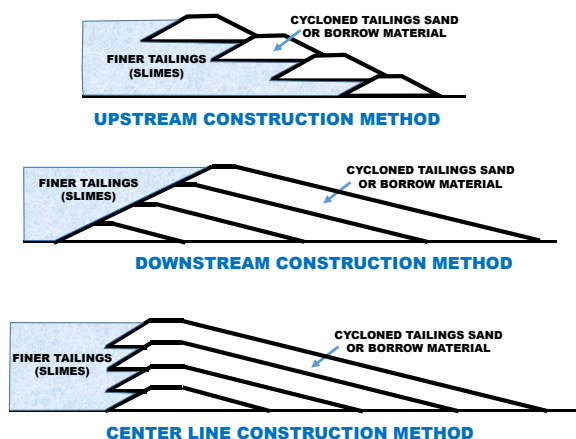


Figure 1. Methods of tailings dam construction. Conventional deposits.

From the resulting geometry of the dams, it is evident that the upstream method is the most questionable in terms of stability because each growth stage is achieved through small dams that rest on the stored slimes that present low strength. In contrast, the downstream construction method has the most stable condition because it generates a dam that is entirely built with selected and controlled material (cycloned tailings sands or borrow material) and is fully supported by a foundation ground. An intermediate situation corresponds to the centerline method of construction, in which the crest of the dam increases vertically, leaving the entire downstream shell built on competent and controlled material but with the upstream section partially supported on the stored tailings.

Because construction costs are directly related to the required volumes of cycloned sands or borrow material, it is apparent from an economic analysis that upstream dams are the most attractive, followed by the center line and then downstream dams. Consequently, around the world, many tailings dams are built by the upstream method due to their lower cost, regardless of the greater risk of their physical stability.

It is essential to mention that the recent tailings dam failures that occurred in Brazil (Fundao and Brumadinho) involved tailings dams built with the upstream method, whose catastrophic collapse is related to flow failures. While the tailings dam Mount Polley that failed in Canada was built using the centerline method with a modification that resulted in a slight upstream construction of the crest. Its failure is mainly attributed to the existence of a weak clay layer in the foundation ground plus insufficient beaches to protect the embankment from the surplus of water once the embankment failed (Hoffman, 2014).

3 CHILEAN EXPERIENCE

Chile is considered a mining country due to its remarkable production of copper, molybdenum, iron, and other metals and non-metallic products as lithium, saltpeter, iodine, among others. The high mining activity, which began several centuries ago, has imposed the need to develop extensive tailings storage facilities, which have to be stable when exposed to strong seismic disturbances according to the high seismicity of the Chilean territory. In this context, Chile has developed an important experience associated with the seismic response of large tailings dams (Valenzuela 2015).

The Chilean seismic environment results from the subduction of the Nazca Plate below the South American Plate, which is converging at an estimated rate of 65 to 80 mm per year. As a result, the largest earthquake recorded in the world, the Valdivia Earthquake of magnitude, $M_w = 9.5$, occurred in the southern part of Chile in 1960. Recently, the sixth earthquake worldwide took place in 2010 in the south-central area of Chile, with a magnitude, $M_w = 8.8$. Historical data indicate that every eight years, an earthquake of magnitude equal to or greater than 7.5 shakes the country. Therefore, there is no room for improper designs; otherwise, failures would be inevitable.

The Chilean history of tailings dam behavior is obscured by two major failures triggered by seismic events. The oldest reported failure corresponds to the Barahona tailings dam that occurred on December 1, 1928, due to the 8.4 Richter Magnitude Talca Earthquake. The Barahona dam that reached 65 m failed and the tailings flowed through the Coya River, killing 54 people, more details presented below. The second failure, which is the most catastrophic in the Chilean experience, occurred due to the La Ligua earthquake of magnitude 7.4 on March 28, 1965. The tailings dam El Cobre 1 collapsed due to liquefaction and the tailings flowed downstream, burying more than 200 people who lived in El Cobre town. The significant loss of life due to an entire town buried under several meters of tailings caused a severe commotion in the country. Consequently, the authorities promulgated in 1970 the first Chilean code that regulated the design, operation and construction of tailings storage facilities, where the construction of the tailings sand dams was only authorized using the downstream or the centerline methods. Accordingly, in Chile, since 1970, law prohibits upstream tailings sand dams.

It is essential to mention that the critical factors of this code are five:

- 1) The construction of the tailings sand dams was only authorized using either the downstream or the centerline methods.
- 2) The tailings sands have to be compacted.
- 3) It introduced the concept of "dangerous distance," equivalent to what today is the breach runoff assessment.
- 4) The stability analysis had to include a seismic coefficient based on the potentially affected population. For example, 0.1 for no population and 0.20 for ten thousand persons.
- 5) The design had to meet a minimum safety factor equal to or greater than 1.2.

This code was significantly positive because no failures have been reported in tailings dams constructed with the downstream or centerline methods since 1970.

Later on, in 2007, a new code was promulgated, currently valid. It maintains essential parts of the previous code, such as the prohibition of the upstream method of construction and the requirement of a pseudo-static factor of safety equal to or greater than 1.2. Among other new requirements, it requests a dynamic analysis of the dams.

Unfortunately, some tailings dams built before the code of 1970, using the upstream method, have failed or undergone severe damages (see Table 1).

After the code issued in 1970, more than 11 earthquakes of $M_w \geq 7.5$ have struck the Chilean territory, the largest being the Maule Earthquake of $M_w = 8.8$. There are three important copper

mines in the affected seismic area of this mega-earthquake: El Teniente, Andina and Los Bronces, with several tailings deposits none of which reported failures or severe damages. This empirical evidence indicates that the tailings dams constructed according to appropriate technical requirements (downstream construction method, compacted tailings sands, factor of safety > 1.2 , plus drains and other details) perform properly even under strong seismic disturbances.

Table 1. Chilean tailings dams that experienced failure after 1970

Tailings Dam	Earthquake	Type of dam
Veta del Agua No1	Valparaíso, 1985	Upstream
Cerro Negro No4	Valparaíso, 1985	Upstream
Veta del Agua	Maule, 2010	Upstream
Bellavista	Maule, 2010	Upstream
Las Palmas	Maule, 2010	Upstream*

* This dam had a second construction on top of the first

It is possible to conclude that the Chilean experience associated with the behavior of TSF is strongly influenced by the failure of the El Cobre tailings dam, which resulted in the first code that regulated the construction of these deposits. Probably, because the construction of dams with the upstream method was prohibited, the empirical outcome is that large tailings deposits are perfectly stable if the usual issues considered in dam design are adopted.

4 BEHAVIOR OF SANDY SOILS

The critical state soil mechanic (CSSM) was initially developed and accepted for silty and clayey materials. After several years of discussions and research, it is possible to indicate that this working framework has also been accepted to be valid for sandy soils, and therefore, applicable to tailings materials. However, still, for sandy soils, some researchers prefer to use terms as steady state or ultimate state to refer to the CSSM.

Casagrande initially postulated the concept of a critical void ratio in 1935 (Casagrande, 1936). Using direct shear tests, Casagrande observed that during shearing, dense sand expands (increasing its void ratio) while very loose sand contracts (reducing its void ratio). Casagrande thought that the critical void ratio or critical density was a unique constant value for a given soil, but later, he realized that the critical void ratio is a function of stress levels. On the other hand, testing steel balls in the simple shear apparatus Roscoe and his co-workers presented a conclusive study proving the concept of the critical void ratio, which was extended to clayey soils (Roscoe et al., 1958). Notwithstanding the foregoing, the first study in sandy soils that showed the drastic and sudden drop of strength that loose sands can experience under undrained loading conditions was reported by Castro (1969). Testing more soils and considering the results of Castro, Poulos (1971) introduced the term "steady state of deformation" to refer to the state of failure of continuous deformation carried on by a particulate material under a constant state of effective stress at constant velocity and constant void ratio. Fig. 2 is illustrated a stress-strain curve of contractive sand subjected to an undrained loading, where Poulos identified three zones:

- Zone A: Shape strongly affected by soil type, initial structure, initial state, and method of loading. Initial structure (including bonds, if any) and state have more influence here than in any other zone.

- Zone B: Initial structure and state increasingly altered by strains until steady state of deformation is reached. Magnitude of

drop from τ_m to τ_s is controlled chiefly by initial state, degree of void ratio change that is allowed, and grain shape. Initial structure and method of loading also influence magnitude of τ_m/τ_s .

- Zone C: At strains beyond point S crushing has stopped and the grains have reached a steady state "structure". The initial structure and the state have been completely altered by the loading process and have no influence on τ_s . Nevertheless, the initial conditions do influence ϵ_s .

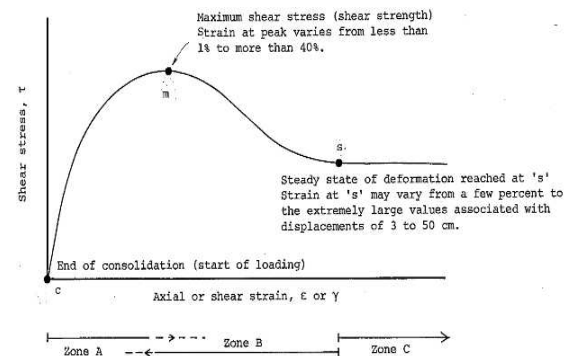


Figure 2. Stress-strain curve of contractive loose sand under undrained load (Poulos 1971).

The steady state concept and its absolute uniqueness for given sand generated important debates among researchers, but when Been and Jefferies (1985) introduced the state parameter ϕ , based on the steady state line in the $e - p'$ plane, the geotechnical community began to be less reactive to the steady state as a fundamental part of the behavior of sands. A comprehensive study using the standardized Japanese sand, Toyoura, and other granular materials, showed the attributes of the steady state line to provide essential information regarding the mechanical behavior of granular materials (Ishihara 1993, Verdugo 1993, Verdugo & Ishihara 1996). The steady state line of Toyoura sand established from triaxial tests is plotted in Fig. 3 in the $e - p'$ plane. In addition, examples of undrained (Fig. 4) and drained (Fig. 5) test results of loose and medium dense states are presented.

These test results show that a unique steady-state line in the $e - p'$ space can be achieved regardless of the initial soil density (loose or dense), type of load (drained or undrained) and level of pressure. Furthermore, in Fig. 6, the test results of two samples consolidated anisotropically, then loaded monotonically, and cyclically are presented. These results confirm that the steady-state strength is not affected by the stress history and is only a function of the void ratio.

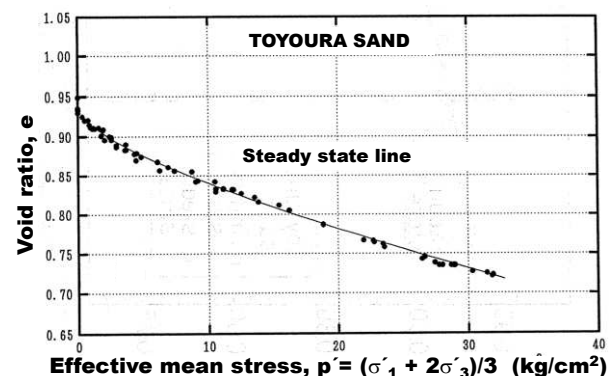


Figure 3. Steady state line in the $e - p'$ plane (Verdugo 1992, Ishihara 1993)

The catastrophic failures due to liquefaction that have experienced tailings dams can be explained in the framework provided by the steady state. In Fig. 7, the results of triaxial tests

on loose samples consolidated anisotropically at different levels of initial shear stress are presented (Castro 1969).

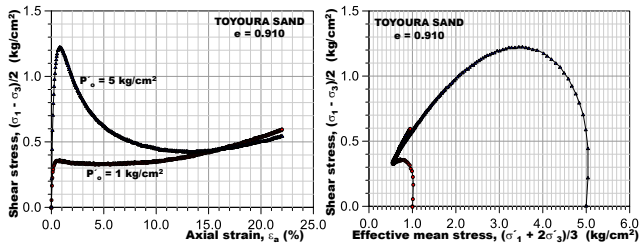


Figure 4. Undrained response at the same void ratio and different confining pressure (Verdugo 1992).

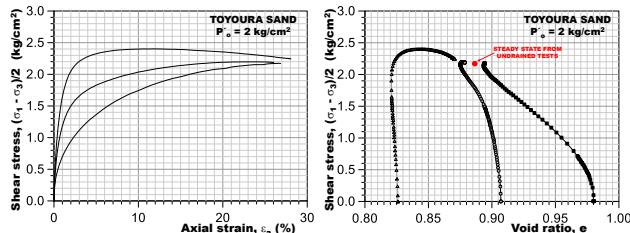


Figure 5. Drained response at the same confining pressure and different initial void ratios (Verdugo 1992).

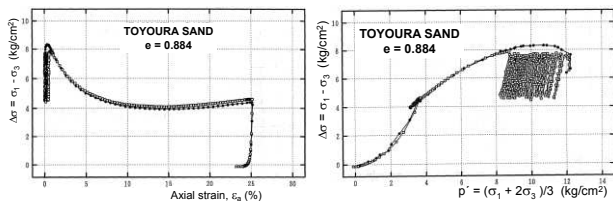


Figure 6. Undrained response of samples with initial static shear stress and with and without cyclic loading (Verdugo 1992).

It is evident that under a drained load condition, it is possible to reach a shear stress level that can be significantly higher than the undrained steady state strength developed at large strains. In addition, the higher the level of the driving shear stress, the lower the stress increment needed to mobilize the maximum resistance. This means that in the case of loose sands subjected to a high level of static (or permanent) shear stress, any minor disturbance can trigger the undrained response of the material and its steady state strength. Similar experimental evidence has been reported by Castro and Poulos (1977) and Kramer and Seed (1988), among others.

Consequently, the failure occurs, and the soil mass should deform and flow until the driving stresses decrease to a value compatible with the residual undrained resistance (steady state strength).

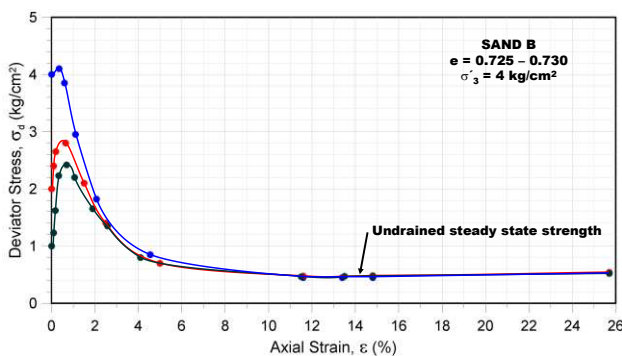


Figure 7. Undrained response of samples initially consolidated at different level of static shear stresses (from Castro 1969).

5 UNDRAINED STRENGTH OF A TAILINGS SAND

To investigate the undrained residual strength of copper tailings sands, a batch of this material was extracted from a Chilean tailings sand dam and tested in the laboratory using the triaxial equipment (Mora 2001; Verdugo 2009). A cycloned tailings sand with 18% of non-plastic fines constitutes the dam. The grain size distribution curves of the washed sand and the fines are presented in Fig. 8. Three different batches of tailings sands were tested: the original sand (18% of fines), the washed sand (0% of fines) and the original sand with an addition of 12% of fines (30% of fines). The maximum and minimum void ratios of each batch are presented in Table 2. The specific gravity of these batches is 2.71. The steady state lines of the tested tailings sand with different fines contents are shown in Fig. 9 in the e-Su plane.

Table 2. Maximum and minimum void ratios of tested tailings sands

Fines Content (%)	e_{max}	e_{min}
0	1.177	0.639
18	1.238	0.589
30	1.277	0.499

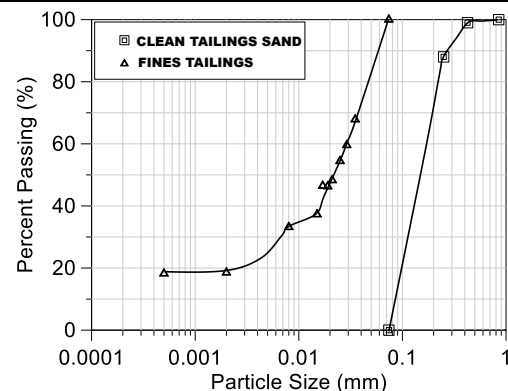


Figure 8. Grain size distribution curves of clean (washed) tailings sand and tailings fines.

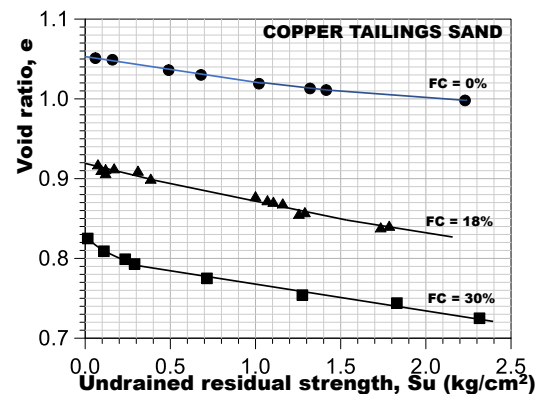


Figure 9. Steady state lines in the e-Su plane of the tailings sand tested with different fines contents

It is observed that the increase of non-plastic fines significantly affects the location of the SSL, reducing the undrained residual strength drastically for any given void ratio. This experimental evidence is highly relevant as it implies that tailings sands strongly increase their potential to develop a flow failure as their fines content increases.

For the mechanical characterization of clayey soils, the importance of the normally consolidated and critical state lines in the e-p' plane is well accepted and recognized. In sands, it is possible to consider the SSL and two additional lines: the

consolidation lines of the loosest and densest states, which are associated with the maximum and minimum void ratios, respectively. The SSLs of the tested tailings in the e - p' plane are shown in Fig. 10, including the estimated consolidation curves for the loosest state (curve parallel to the SSL that starts at $e_{\max}$) and densest state ($e = e_{\min}$ line). These approximations do not change the essential empirical evidence that shows that as the content of fines increases, the area of possible contractive states increases significantly.

The parameter Relative Contractiveness, R_c , (Verdugo & Ishihara, 1996), allows to estimate how contractive a given soil is; R_c close to zero represents soils with a small contractive zone in the e - p' plane. On the contrary, R_c close to one is associated with soils with a large contractive zone.

The computed values of R_c , considering the approximations indicated, are 0.24 (0 %FC), 0.52 (18%FC) and 0.63 (30%FC). Therefore, it is evident that the non-plastic fines increase the intrinsic contractiveness of this copper tailings sand. This result has important practical impacts:

- 1) When cycloned sands are used to build a tailings sand dam, the lower the fines content, the better their mechanical response.
- 2) Tailings sands with high fines contents need much higher efforts of compaction to avoid contractive response.

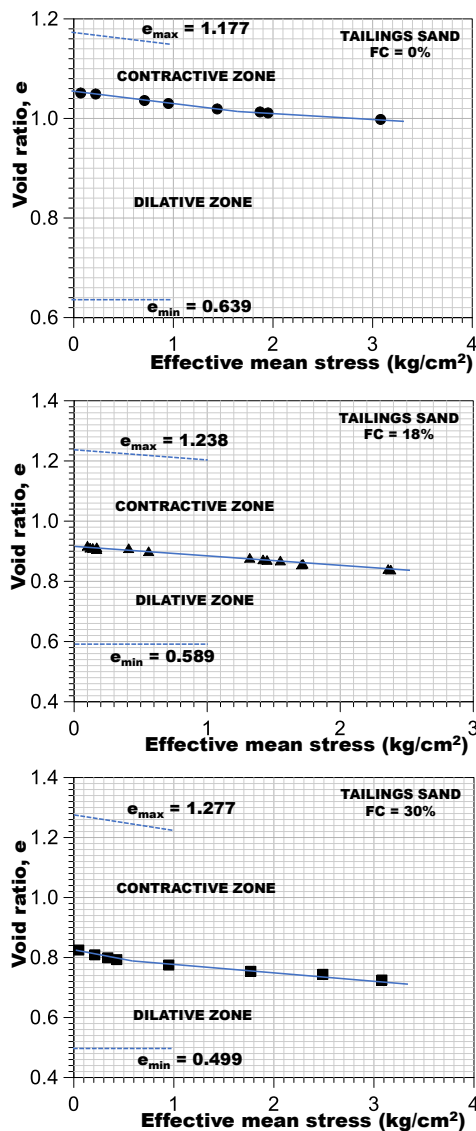


Figure 10. Steady state lines in the e - p' plane and the zones of contractive and dilative response.

6 REVISITING THE BARAHONA DAM FAILURE

At 0 hours 6 minutes on December 1, 1928, an earthquake of Richter Magnitude 8.4 struck the central-south part of Chile (Lomnitz 2004). During this earthquake, an engineer in charge of the construction of the Barahona dam witnessed that the dam remained stable, since the lighting on the dam was visible. However, two or three minutes after the shaking ended, the electricity was cut off and the telephone did not work due to the failure of the line that passed about 200-300 m downstream of the dam (Aguero 1929).

Initially, this TSF was designed with a dam built from borrow materials using the downstream construction procedure. However, due to some construction difficulties and high cost, the project was modified, and the construction of an upstream tailings sand dam was implemented. This deposit began receiving tailings from the El Teniente mine on March 31, 1920. The reported fines content of the tailings sands used in the dykes was around 20%, while the reported fines content of the slimes stored in the pond was in the range of 83 to 92%.

At the time of the earthquake and subsequent failure of a sector of the dam, the maximum height of the dam was 65 m, the upstream slope had an overall inclination of 2:1 (H:V), a freeboard of 17 m and a crest width of 12 m, as is shown in Fig. 11. The dam had a total length of 1,885 m, the failure occurred in the highest part of the dam, and consisted of a breach of 440 m in length, which compromised the total height of the dam. The geometry of the surface of the tailings that remained in the zone of the failure developed a stepped shape; rather flat terraces and almost vertical scarps were observed (Fig. 12 Troncoso et al, 1993).

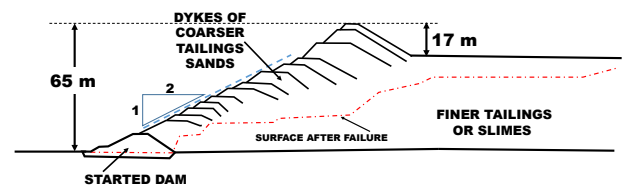


Figure 11. Cross section of Barahona tailings sand dam and surface left after failure (source of information: Aguero 1929).



Figure 12. Geometry of the failure surface (Troncoso et al, 1993)

According to Agüero, approximately 27 million tons of tailings were stored at the time of the failure, of which about 4 million tons flowed out of the reservoir. The tailings flow went down violently through the Barahona Ravine and then reached the Coya River. In the zones where the ravine changes its direction, the tailings flow reached a height of 60 to 70 m. Small hamlets on the banks of the Coya River were totally destroyed, causing the death of 54 people.

The Barahona Dam failure occurred a few minutes after the earthquake, so no inertial forces were acting on the dam at the time of the failure. The tailings flowed out suddenly, indicating their liquefaction. Under these considerations, a limit equilibrium analysis of the Barahona Dam was performed assuming for the stored tailings a normalized undrained residual

strength of 0.10, and for the tailings sand dykes a drained response considering a friction angle, $\phi = 30^\circ$ and no cohesion. The results confirm the failure of the deposit; however, the failure surfaces resulting from the numerical analysis differ from the stepped form in which the tailings were left in the impoundment. Therefore, the observed surface of the tailings remaining after failure is likely to be affected by the erosion process associated with the flow of tailings and water. This is an important issue to take into account when back analysis are performed.

7 CONCLUSIONS

The review of the Chilean experience in the design, construction and operation of tailings storage facilities, indicates that the catastrophic failure of El Cobre tailings dam, which occurred in 1965, generated a significant change in the Chilean legislation, which resulted in the prohibition of the upstream construction method of tailings sand dams. Besides other important aspects, it is possible to point out that the abolition of the upstream construction method is the key factor that has permitted that the strong earthquakes that have hit Chilean territory have not induced failures in the large tailings dams existing in this country.

The occurrence of the liquefaction phenomenon is one of the main factors responsible for some of the large failures that have experienced tailings sand dams. The framework provided by the steady state of deformation can be used to analyze the occurrence of this phenomenon and to estimate the ultimate strength mobilized when the liquefaction is developed. The steady state refers to the state of failure of continuous deformation carried on by a particulate material under a constant state of effective stress at constant velocity and at constant void ratio. For practical purposes, it is equivalent to the Critical State commonly used in fine soils. A comprehensive series of experimental results of sandy soils were presented showing the validity of the steady state.

A Chilean copper tailings sand with different content of non-plastic fines (0, 18 and 30 %) was tested using the triaxial equipment, in order to establish the steady state in the e - q - p' space. The results showed that the higher the content of fines in the sand the more contractive results the sand. At similar void ratio, the fines reduced the residual undrained strength. These experimental results suggest that when cycloned sands are used in a dam at similar compaction, the lower the fines content the better the mechanical behavior.

One of the oldest reported tailings dam failures in Chile occurred at El Cobre dam in 1928, triggered by an earthquake. The rupture was sudden and occurred about three minutes after the earthquake ended. Using undrained residual strength values obtained from different Chilean dams, a numerical analysis clearly indicates the failure of this tailings dam. However, the failure surfaces given by the analysis are different from the stepped geometry of the remaining tailings. It is possible to expect that the erosion generated by the flow of tailings and water controls the final surface observed in the field. Therefore, the observed final surface should be different from the initial failure surface. This situation must be taken into account when performing a back-analysis.

8 ACKNOWLEDGEMENTS

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